

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Touch Display
Manufacturer: C Touch
Firmware Version: 4.2.1
Model(s): Laser, Laser air, Leddura, Leddura xt, Runner

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.03.0000-b006	2.01.0001-b001	1.1.1

Version History

Module Version	Date	Notes
1_0_1_0	12/7/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5
- DeviceID variable must be set accordingly. Default value is 1. DeviceID ranges from 'Broadcast', '1' to '256'. Example: InterfaceName.DeviceID = '1'
- For Serial Over Ethernet support, SerialOverEthernetClass must be used. Example: InterfaceName.SerialOverEthernetClass('192.168.254.254', PortNum, Model='Laser')

Control Commands, States, and Qualifiers:

AspectRatio	None		
AudioMute	'On'	'Off'	
ClosedCaption	None		
Freeze	None		
Input	'TV'	'AV1'	'S-Video 1'
	'Component 1'	'VGA'	'HDMI1'
	'HDMI2'	'HDMI3'	'Display Port'
	'Inside PC'	'UHD'	'USB'
Input_LaserAir	'AV1'	'VGA'	'HDMI1'
	'HDMI2'	'HDMI3'	'USB'
Keypad	'0' – '9'		
MenuNavigation	'Menu'	'Exit'	'Up'
	'Down'	'Right'	'Left'
	'Enter'	'Guide'	
OnScreenDisplay	None		
Power	'On'	'Off'	
TVChannel	'Up'	'Down'	
Volume	0 to 100 in steps of 1		
Volume_LaserAir	0 to 100 in steps of 1		

Status Available:

AudioMute	'On'	'Off'	
ConnectionStatus	'Connected'	'Disconnected'	
Input	'TV'	'AV1'	'S-Video 1'
	'Component 1'	'VGA'	'HDMI1'
	'HDMI2'	'HDMI3'	'Display Port'
	'Inside PC'	'UHD'	'USB'
Input_LaserAir	'AV1'	'VGA'	'HDMI1'
	'HDMI2'	'HDMI3'	'USB'
Power	'On'	'Off'	
Volume	0 to 100 in steps of 1		
Volume_LaserAir	0 to 100 in steps of 1		

Cable and Adapter Requirements:

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device:

Default baud rate for Laser Air model is 115200.

Serial communication:

Port Type: RS-232

Parity: None

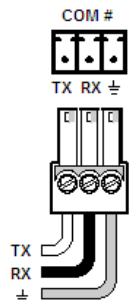
Baud Rate: 115200

Stop Bits: One

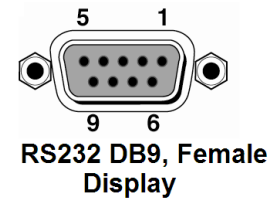
Data Bits: 8

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



General Notes:

Appendix A. Set Commands

Aspect Ratio None	\xA9\x16\x01\x81\x8A
Audio Mute Off	\xA9\x14\x01\x00\x8A
Audio Mute On	\xA9\x14\x01\x01\x8A
Closed Caption None	\xA9\x16\x01\x82\x8A
Freeze None	\xA9\x16\x01\xB8\x8A
Input AV1	\xA9\x15\x01\x01\x8A
Input Component 1	\xA9\x15\x01\x03\x8A
Input Display Port	\xA9\x15\x01\x08\x8A
Input HDMI1	\xA9\x15\x01\x05\x8A
Input HDMI2	\xA9\x15\x01\x06\x8A
Input HDMI3	\xA9\x15\x01\x07\x8A
Input Inside PC	\xA9\x15\x01\x09\x8A
Input S-Video 1	\xA9\x15\x01\x02\x8A
Input TV	\xA9\x15\x01\x00\x8A
Input UHD	\xA9\x15\x01\x0A\x8A
Input USB	\xA9\x15\x01\x0B\x8A
Input VGA	\xA9\x15\x01\x04\x8A
Keypad 0	\xA9\x16\x01\xD9\x8A
Keypad 1	\xA9\x16\x01\x95\x8A
Keypad 2	\xA9\x16\x01\x99\x8A
Keypad 3	\xA9\x16\x01\x9D\x8A
Keypad 4	\xA9\x16\x01\xD6\x8A
Keypad 5	\xA9\x16\x01\xDA\x8A
Keypad 6	\xA9\x16\x01\xDE\x8A
Keypad 7	\xA9\x16\x01\x96\x8A
Keypad 8	\xA9\x16\x01\x9A\x8A
Keypad 9	\xA9\x16\x01\x9E\x8A
Menu Navigation Down	\xA9\x16\x01\xD8\x8A
Menu Navigation Enter	\xA9\x16\x01\x9B\x8A
Menu Navigation Exit	\xA9\x16\x01\xD4\x8A
Menu Navigation Guide	\xA9\x16\x01\xDB\x8A
Menu Navigation Left	\xA9\x16\x01\x97\x8A
Menu Navigation Menu	\xA9\x16\x01\x84\x8A
Menu Navigation Right	\xA9\x16\x01\x9F\x8A
Menu Navigation Up	\xA9\x16\x01\x92\x8A
On Screen Display None	\xA9\x16\x01\xB7\x8A
Power Off	\xA9\x11\x01\x00\x8A
Power On	\xA9\x11\x01\x01\x8A
TV Channel Down	\xA9\x16\x01\x8E\x8A
TV Channel Up	\xA9\x16\x01\x80\x8A
Volume 0	\xA9\x13\x01\x00\x8A

Appendix B. Query Commands

Audio Mute	\xA9\x14\x01\xAA\x8A
Input	\xA9\x15\x01\xAA\x8A
Power	\xA9\x11\x01\xAA\x8A
Volume	\xA9\x13\x01\xAA\x8A